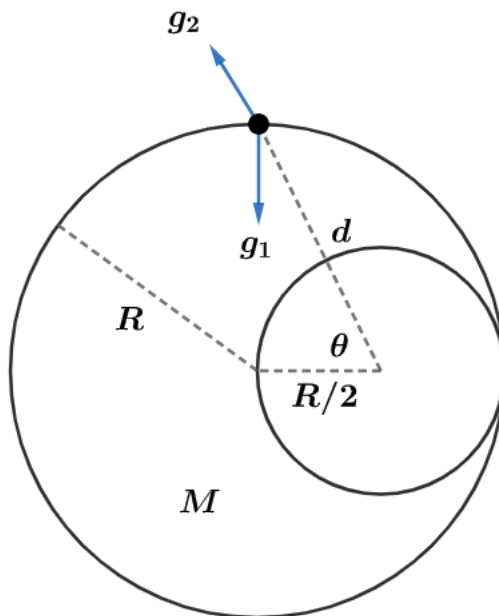


2021 F=ma Exam: Problem 19

Kevin S. Huang



By the principle of superposition, the gravitational field at the top is the sum of the field from the filled spherical planet and the field from treating the cavity as having negative mass. Thus, we have

$$|\vec{g}_1| = \frac{GM}{R^2}$$

$$|\vec{g}_2| = \frac{G(M/8)}{d^2} = \frac{GM/8}{5R^2/4} = \frac{GM}{10R^2}$$

since $d^2 = R^2 + (R/2)^2$. Using the fact that $\cos \theta = 1/\sqrt{5}$, $\sin \theta = 2/\sqrt{5}$, the total field is

$$\begin{aligned} \vec{g} &= \vec{g}_1 + \vec{g}_2 = \left(-\frac{GM}{R^2} \hat{y} \right) + \left(-\frac{GM}{10\sqrt{5}R^2} \hat{x} + \frac{GM}{5\sqrt{5}R^2} \hat{y} \right) \\ &= \frac{GM}{R^2} \left(-\frac{1}{10\sqrt{5}}, \frac{1}{5\sqrt{5}} - 1 \right) \\ |\vec{g}| &= \frac{GM}{R^2} \sqrt{\left(\frac{1}{10\sqrt{5}} \right)^2 + \left(\frac{1}{5\sqrt{5}} - 1 \right)^2} = 0.912 \frac{GM}{R^2} \end{aligned}$$

so the answer is E.